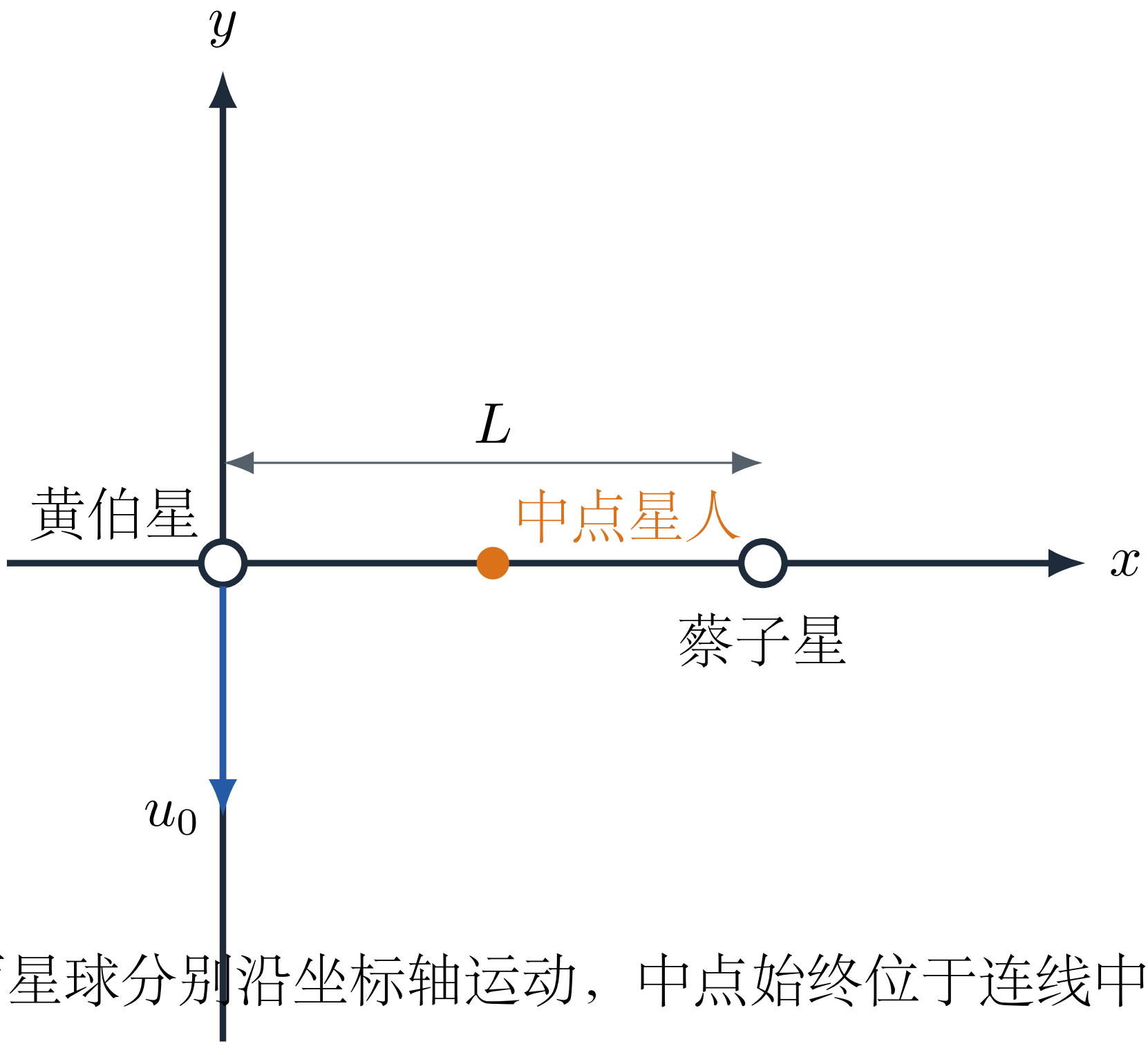
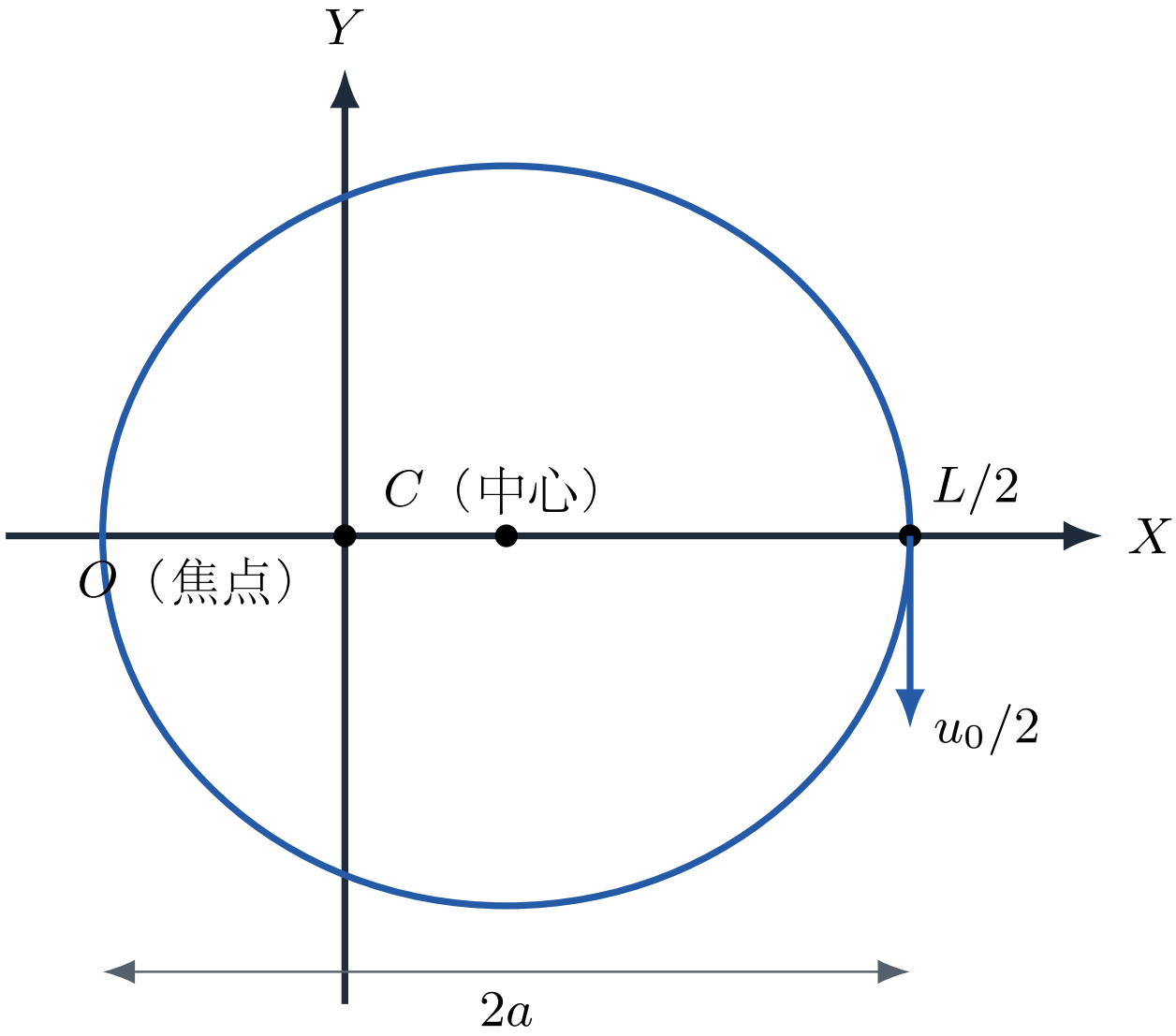


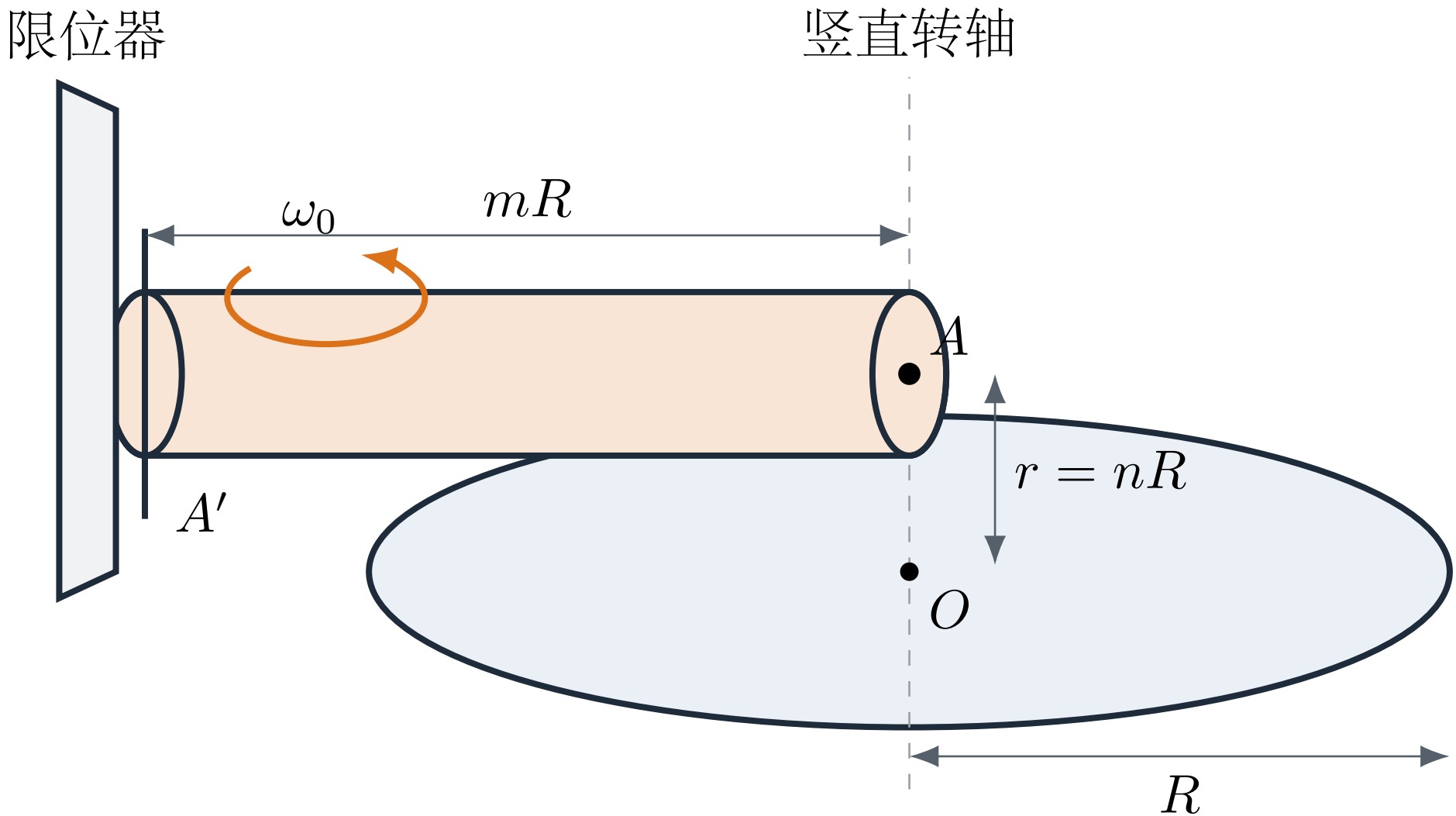
两星球分别沿坐标轴运动，中点始终位于连线中点



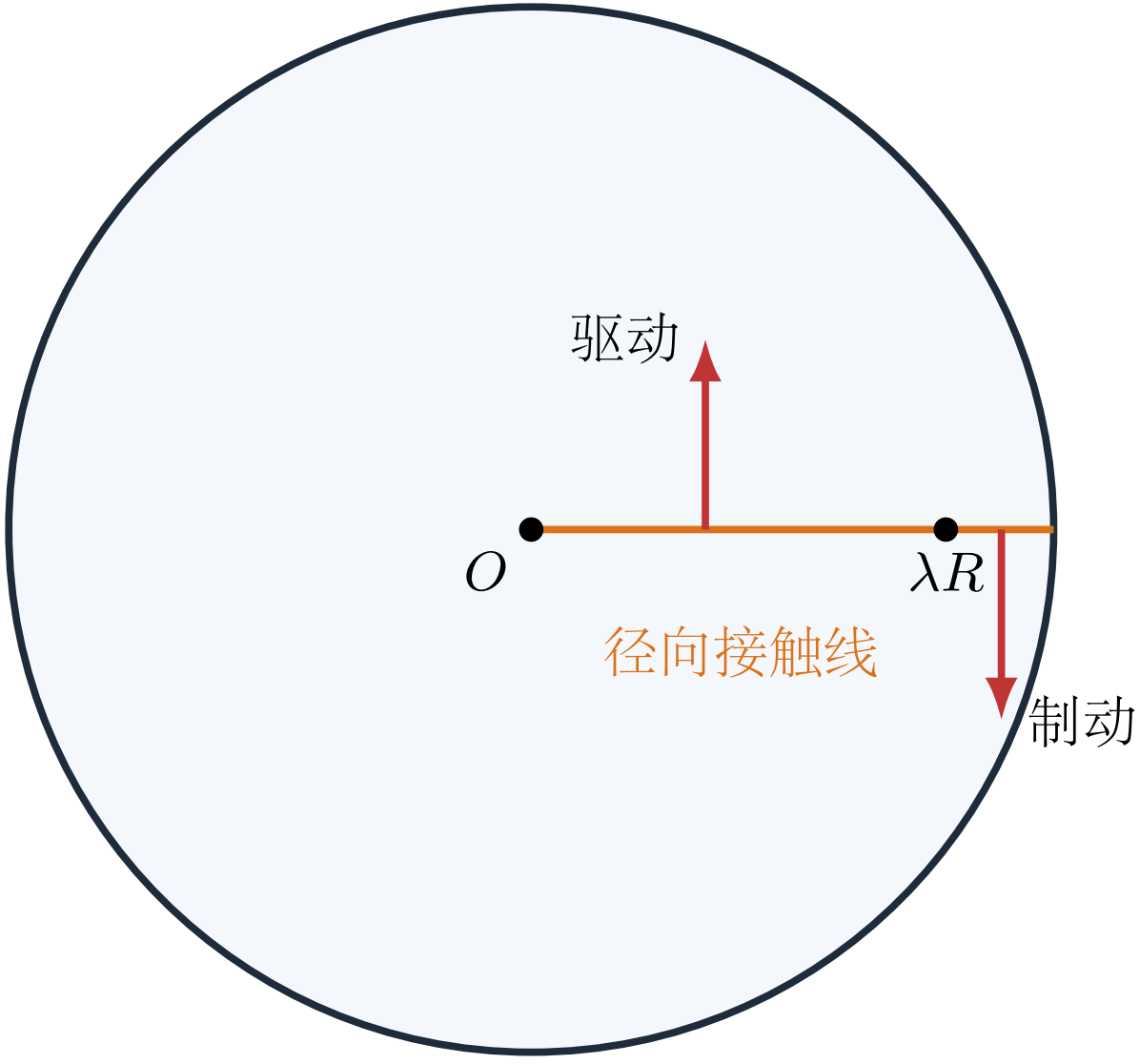
示例 $u_0^2 = 0.6Gm/L$

$a = 5L/14, c = L/7$

$M_{\text{eff}} = m/8$



圆盘绕竖直轴转动；滚筒中心轴为 AA'



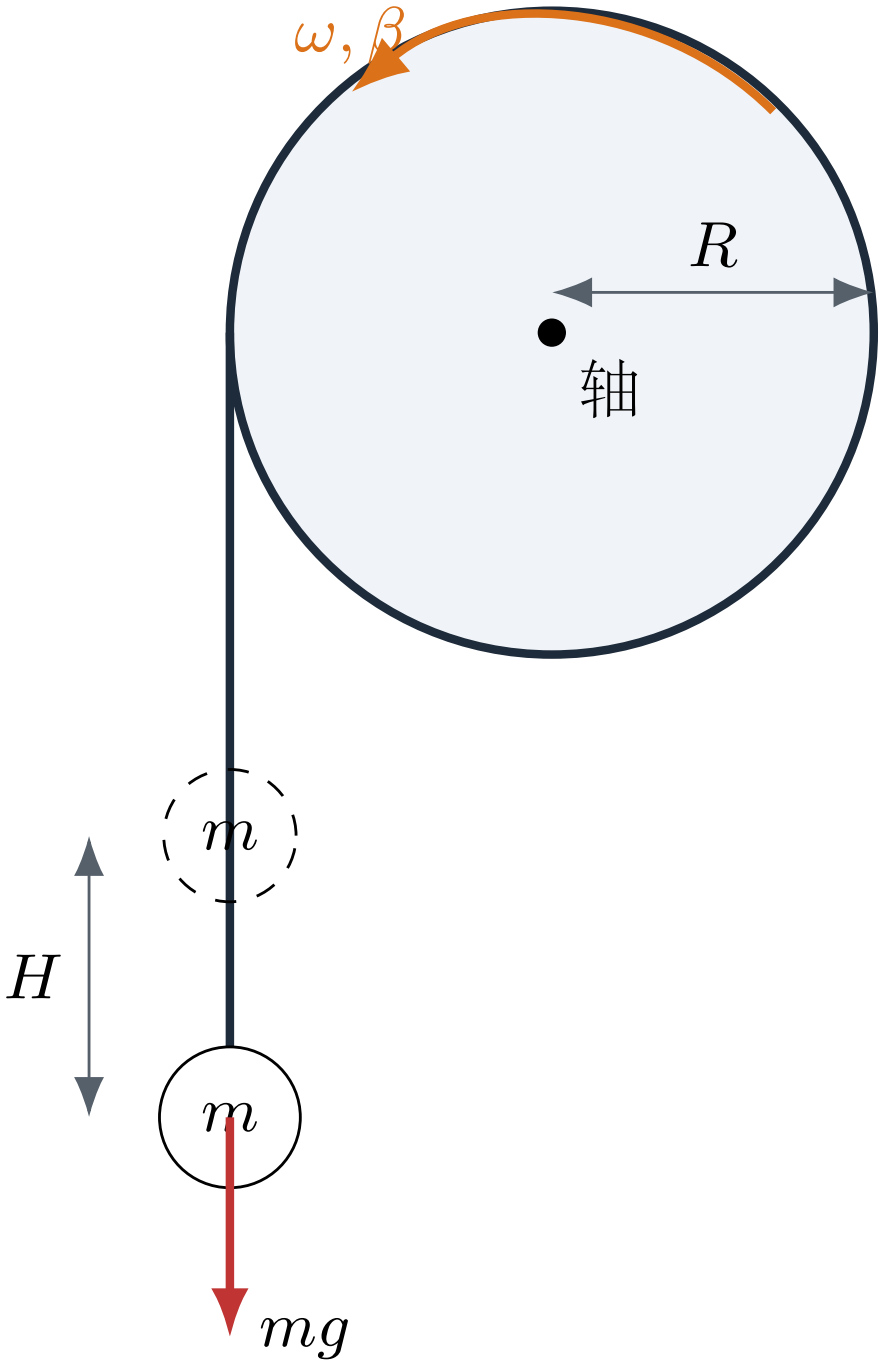
$$\lambda = 2^{-1/3}$$

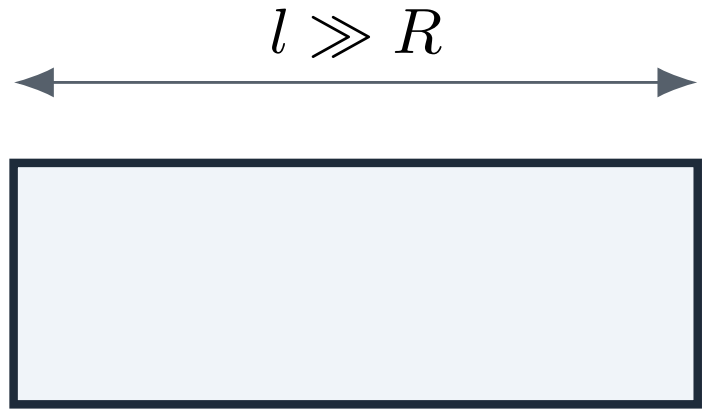
$$\omega_1 \lambda R = \omega_0 n R$$

$$\int_0^{\lambda} \rho^2 \mathrm{d}\rho = \int_{\lambda}^{1} \rho^2 \mathrm{d}\rho$$

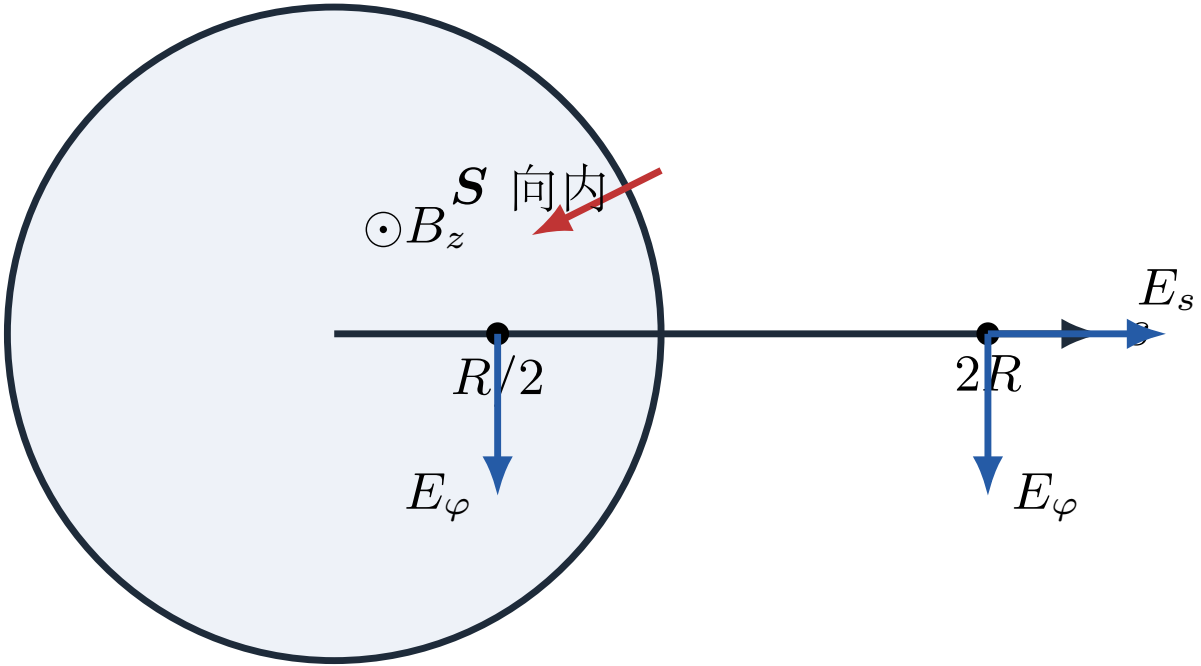
只有接触线上的一个点无相对滑动

端面：轻绳沿左外缘相切





轴向视图：均匀带电 Q



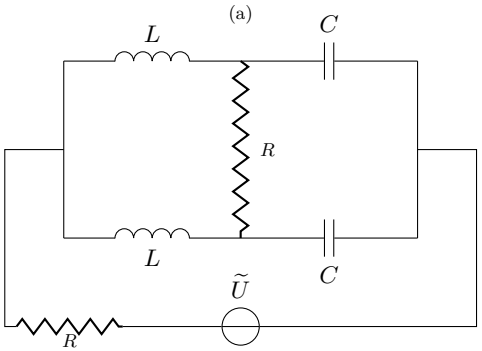
$E_s = 0\ (s < R)$

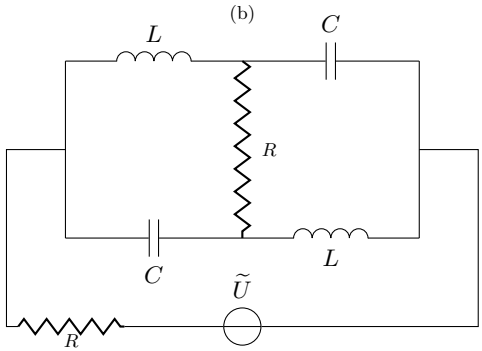
$E_s = 2kQ/(ls)\ (s > R)$

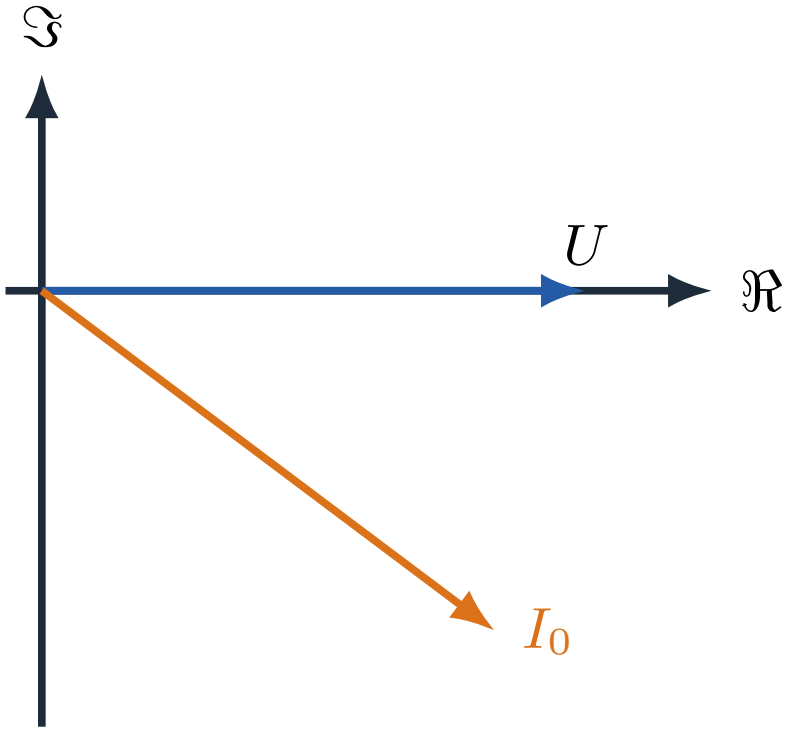
$E_\varphi = -\dot{B}s/2\ (s < R)$

$E_\varphi = -\dot{B}R^2/(2s)\ (s > R)$

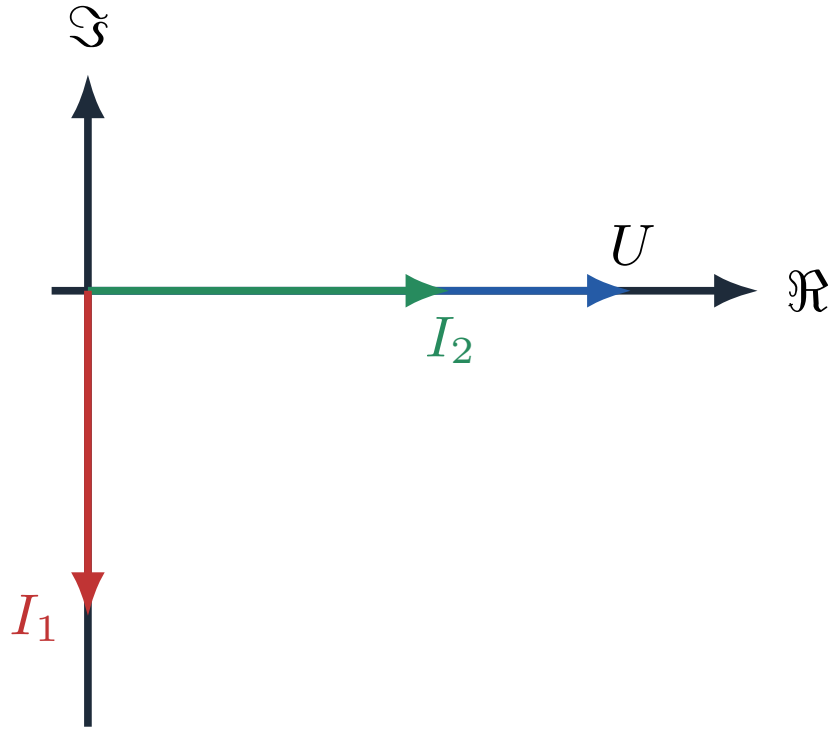
总电场 $\sqrt{E_s^2 + E_\varphi^2}$


(a)
L
C
R
L
C
$\widetilde{U}$
R

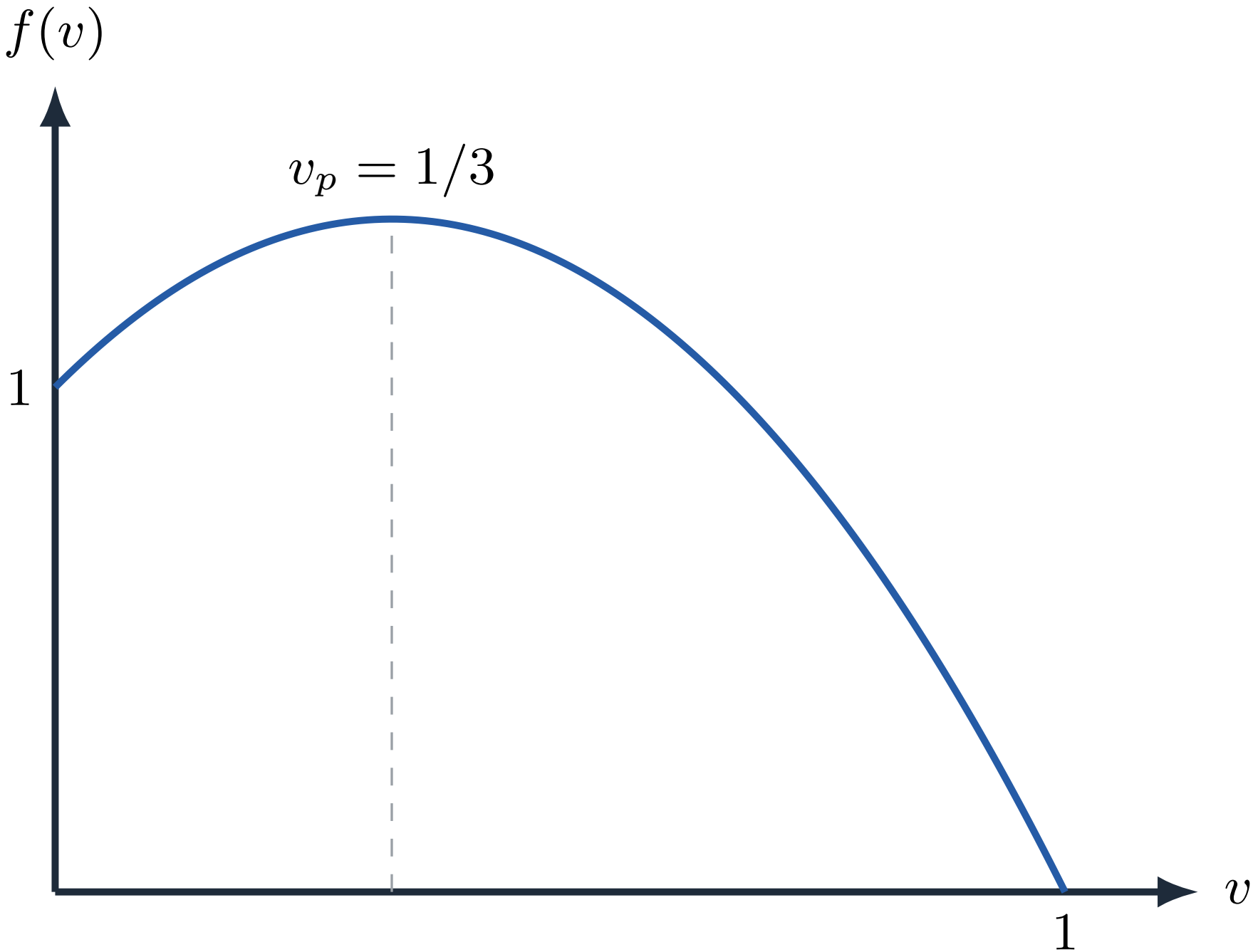

(b)
L
C
R
C
L
$\widetilde{U}$
R

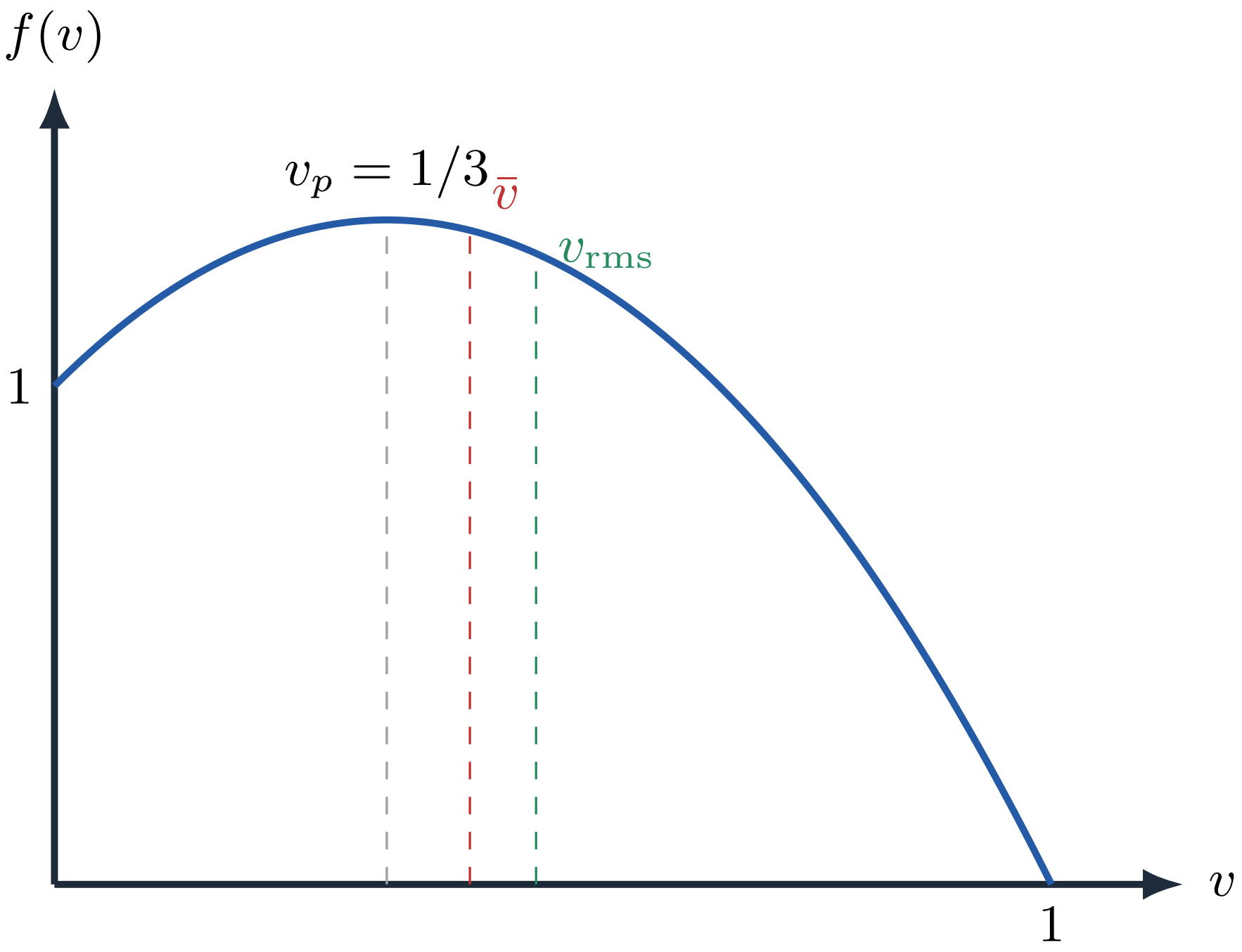


(a) $Z_a = R + 3ir/4$

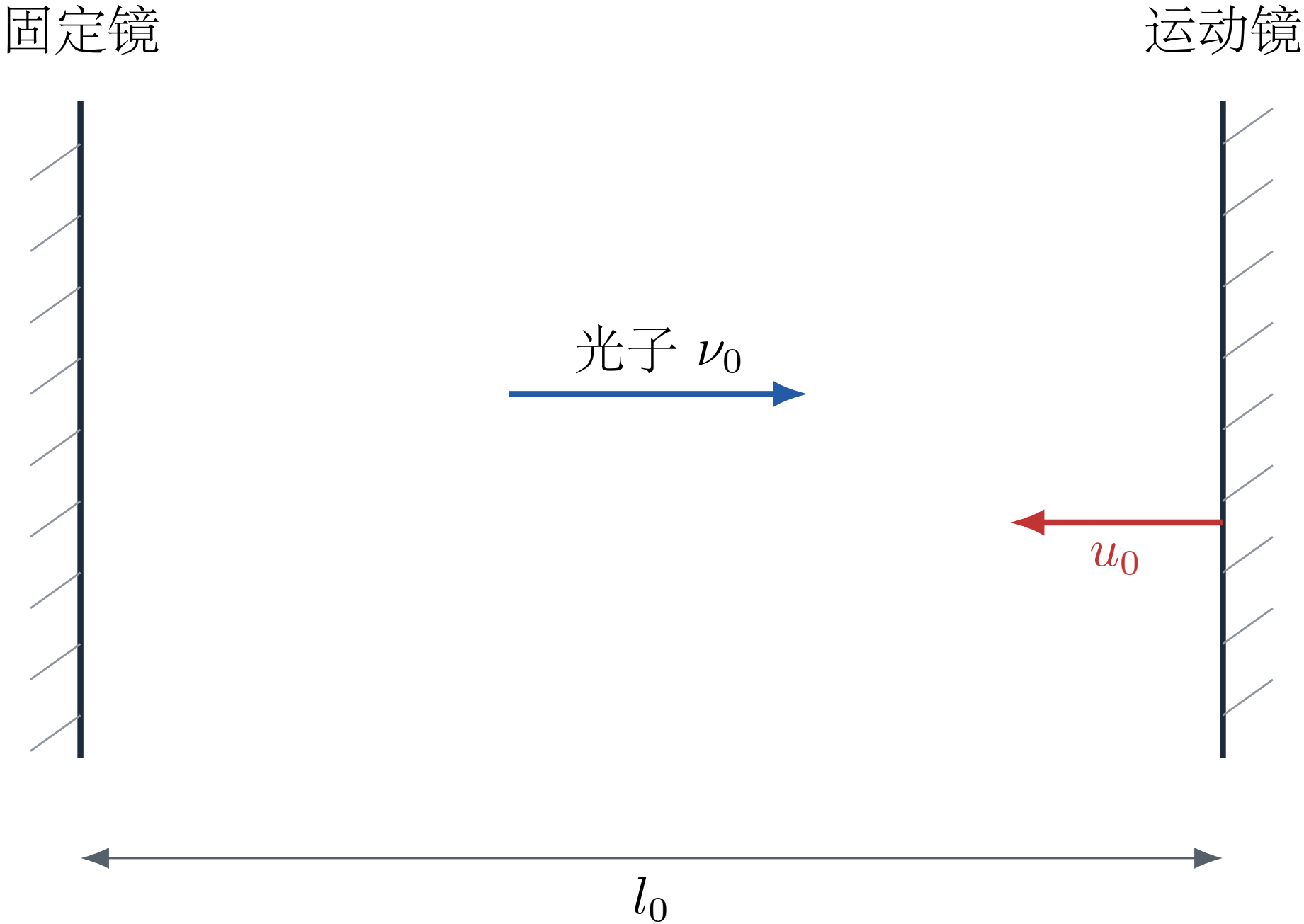


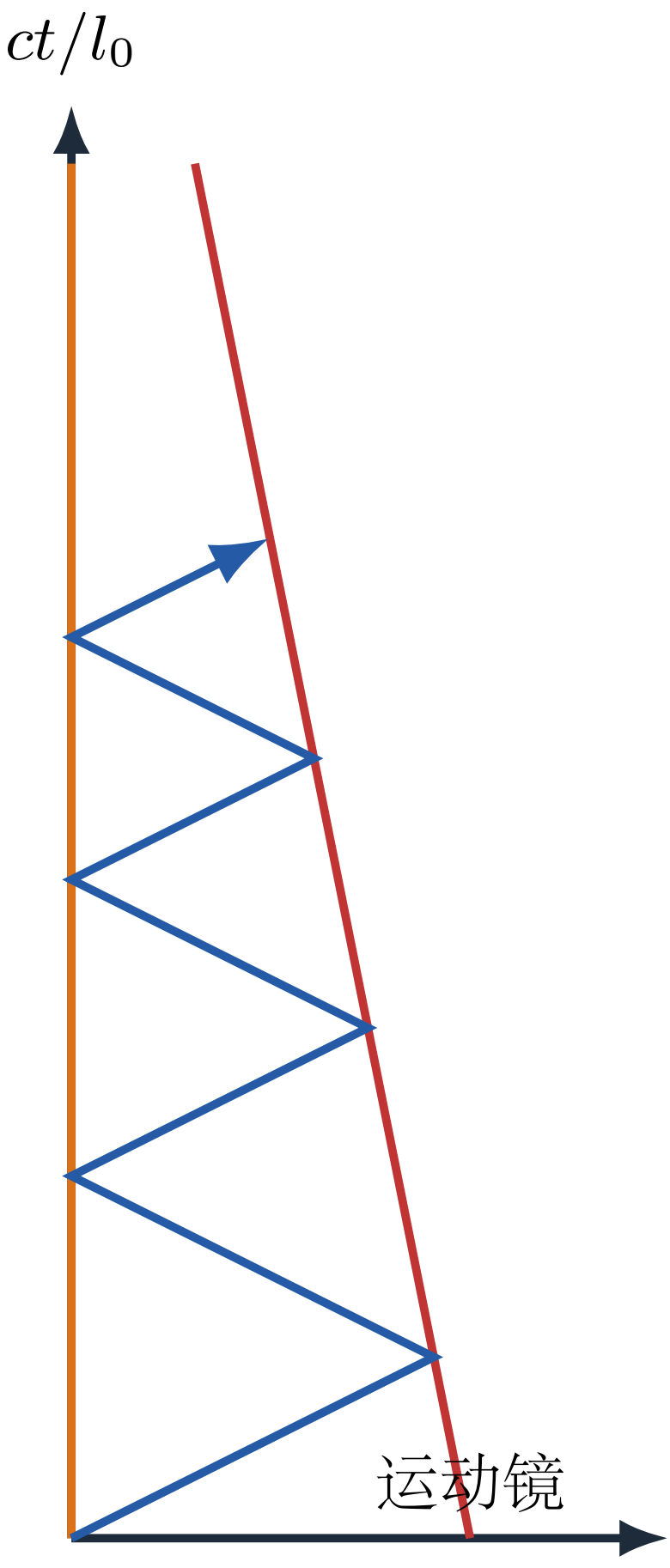
(b) $Z_b = R + r^2/R$


$f(v)$
$v_p = 1/3$
1
v
1



$$\int_0^1 f(v)\mathrm{d}v = 1$$


固定镜
运动镜
光子 ν_0
u_0
l_0

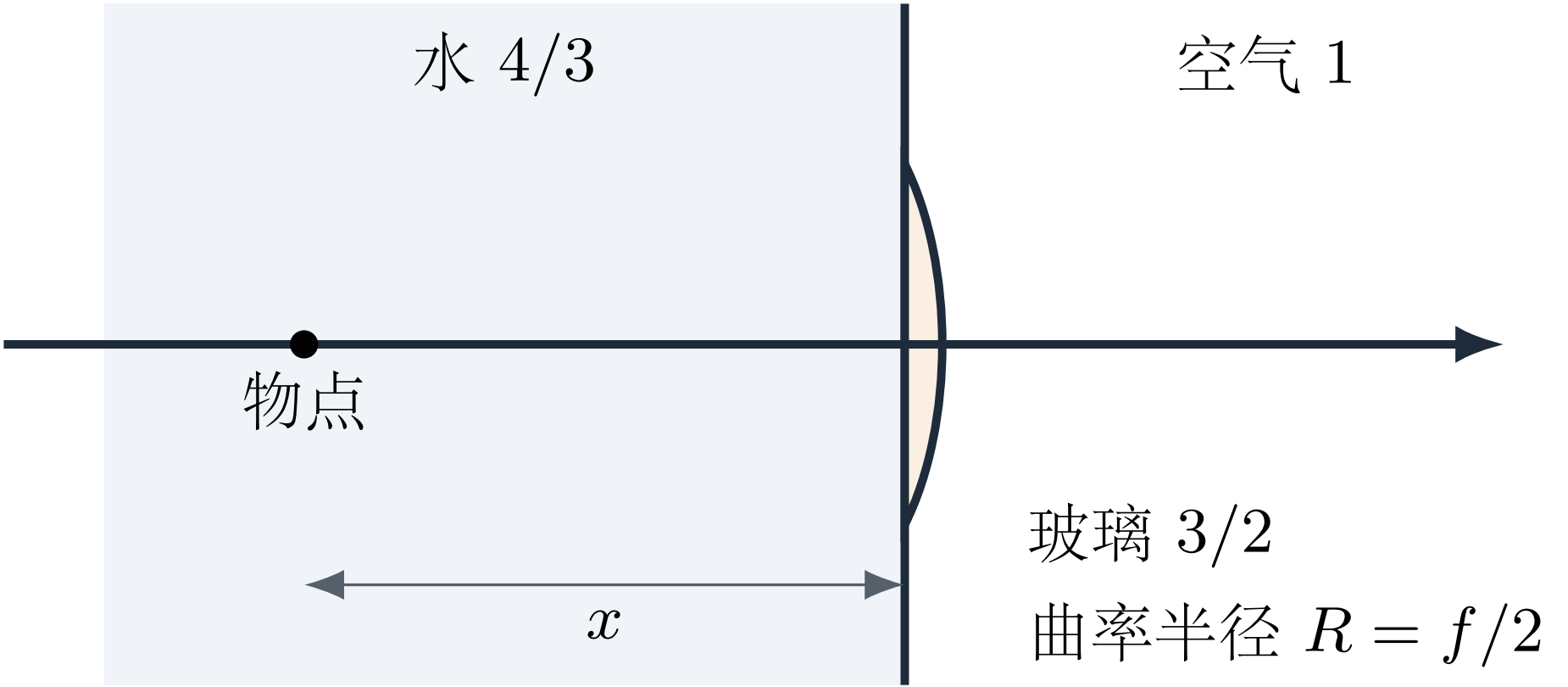


示例 $u_0 = 0.1c$

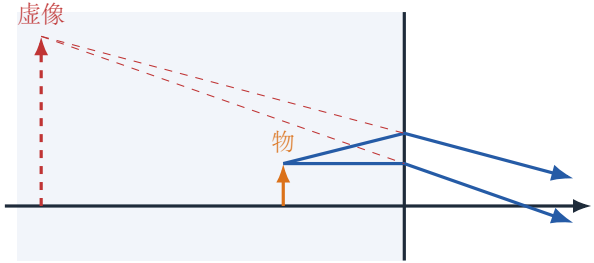
$X_{N+1} = X_N/q$

$\nu_{N+1} = q\nu_N$

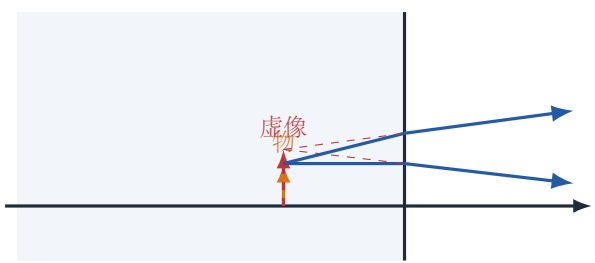
同相位 $\nu_N X_N$ 守恒



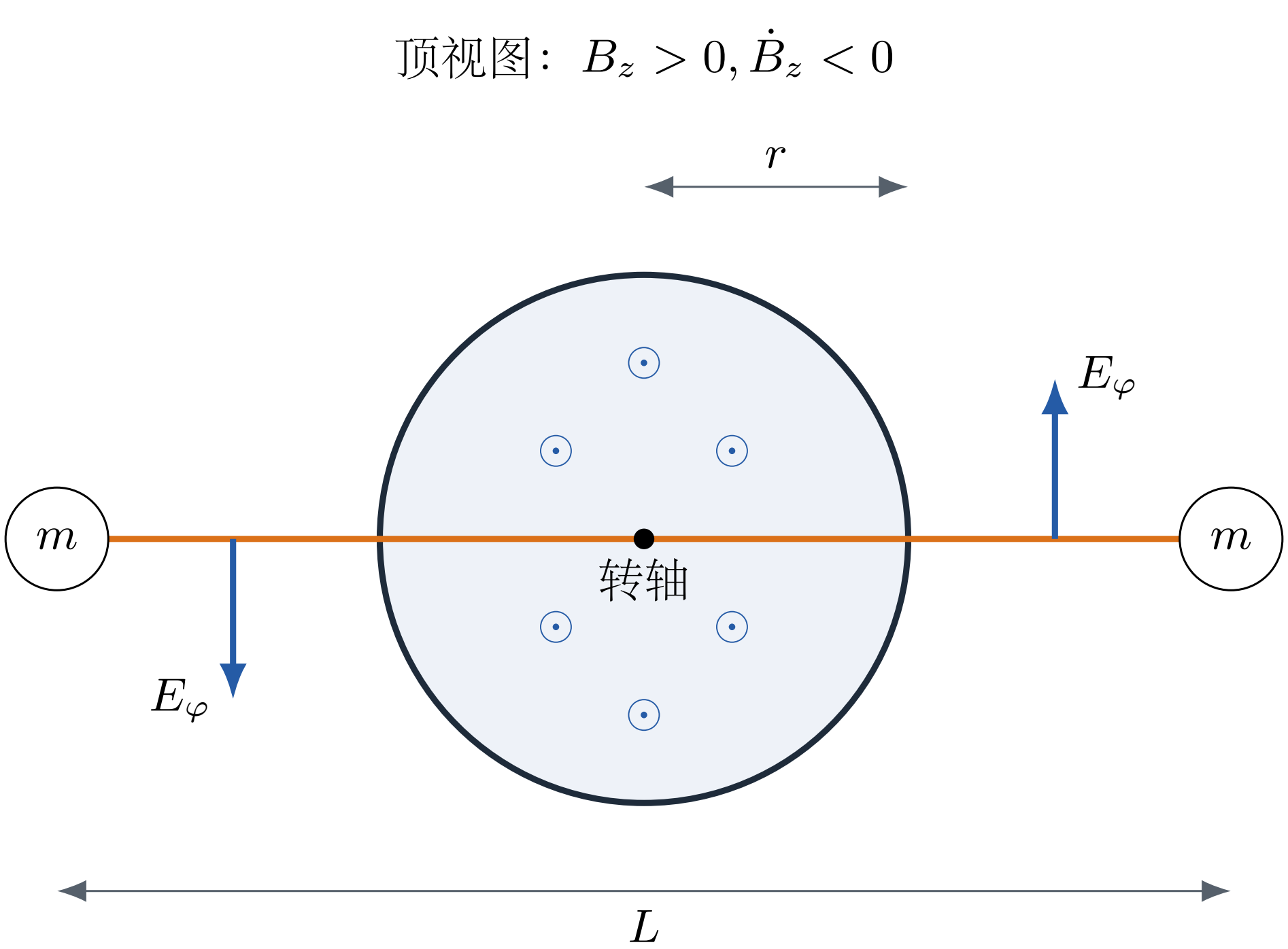
透镜厚度夸张示意；R 是曲率半径

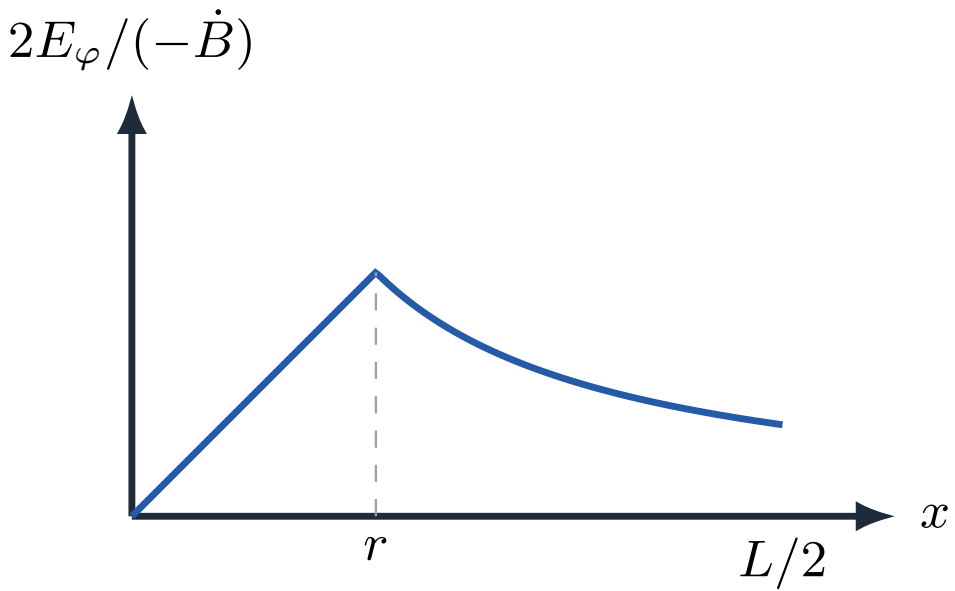


外贴：$y=-3f,\beta=4$

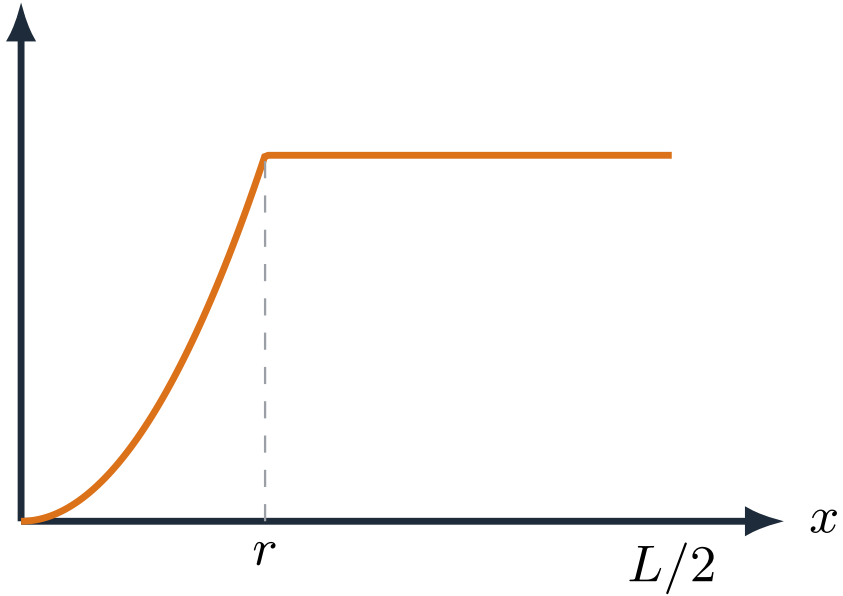


内贴：$y\prime=-f,\beta\prime=4/3$


顶视图：$B_z > 0, \dot{B}_z < 0$
r
E_φ
m
转轴
m
E_φ
L



力矩积分核



x^2 内区，r^2 外区